

WALLA WALLA, WA, USA

WMO: 727846

Lat: 46.095N

Lon: 118.287W

Elev: 1166

StdP: 14.09

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24160

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.2	18.2	5.3	7.5	14.2	11.5	10.3	21.1	32.0	54.0	28.0	52.1	4.8	180	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	98.5	66.1	94.8	64.7	91.1	63.6	68.3	92.6	66.4	90.1	64.9	87.5	8.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	81.1	73.8	57.5	73.6	72.9	55.5	68.5	72.4	33.1	92.8	31.6	90.5	30.4	87.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.1	17.8		9.3	105.0	9.5	3.6	2.4	107.6	-3.1	109.7	-8.4	111.7	-15.3	114.3
				8.5	72.2	9.1	3.8	1.9	74.9	-3.4	77.2	-8.5	79.3	-15.1	82.1
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	36.6	39.7	46.9	52.6	60.5	67.3	77.0	75.6	66.5	53.8	42.6	35.4	(6)
	DBStd	16.04	9.59	8.45	6.64	6.01	7.16	6.88	6.28	6.09	6.75	6.72	8.76	9.49	(7)
	HDD50	1637	421	295	136	36	4	0	0	0	1	36	247	461	(8)
	HDD65	4762	880	708	561	374	180	50	2	3	61	352	672	919	(9)
	CDD50	3327	6	8	40	114	328	518	838	794	496	153	25	7	(10)
	CDD65	975	0	0	0	2	39	118	374	332	106	4	0	0	(11)
	CDH74	11358	0	0	3	41	456	1366	4611	3770	1069	41	1	0	(12)
	CDH80	5461	0	0	0	4	140	554	2482	1886	389	6	0	0	(13)
(14) Wind	WSAvg	8.1	7.2	7.8	9.0	9.0	8.5	8.7	8.5	8.2	7.5	7.4	7.6	7.3	(14)
(15) Precipitation	PrecAvg	19.7	2.2	1.8	2.0	1.7	1.8	1.1	0.7	0.8	0.9	1.7	2.6	2.4	(15)
	PrecMax	28.3	6.9	4.1	4.6	3.9	6.6	3.3	4.7	3.3	2.3	4.1	4.9	5.1	(16)
	PrecMin	12.9	0.6	0.2	0.4	0.2	0.3	0.2	0.0	0.0	0.0	0.0	0.4	0.4	(17)
	PrecStd	3.7	1.3	0.9	1.0	0.9	1.4	0.9	0.9	0.9	0.7	1.1	1.2	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	63.7	71.1	79.0	89.3	97.3	103.9	101.6	93.4	79.3	67.9	62.9	(19)
		MCWB	49.7	49.4	54.0	60.1	64.5	67.5	67.5	66.0	64.5	59.3	55.4	51.0	(20)
	2%	DB	56.4	57.8	65.4	73.4	84.1	90.9	99.4	97.5	88.6	72.8	62.4	57.0	(21)
		MCWB	47.1	48.8	51.6	57.0	62.7	64.8	66.8	65.5	62.5	56.4	51.1	48.2	(22)
	5%	DB	53.1	54.3	61.5	68.8	79.2	86.4	96.3	93.5	84.4	68.3	58.4	52.8	(23)
		MCWB	46.0	46.8	49.9	54.3	59.9	62.7	65.1	63.9	61.1	54.6	50.0	46.0	(24)
	10%	DB	49.6	51.2	58.0	64.2	74.4	81.9	92.7	90.2	80.7	64.4	54.6	48.7	(25)
		MCWB	43.8	44.6	48.0	51.8	57.8	61.0	64.1	62.9	59.4	52.9	47.7	43.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	54.0	56.0	61.9	66.7	70.0	73.0	69.0	67.2	61.3	58.1	52.0	(27)
		MCDB	57.7	58.4	66.8	75.5	84.5	91.3	96.2	96.2	86.8	72.8	65.3	60.7	(28)
	2%	WB	48.5	50.1	53.1	58.0	63.8	66.7	68.8	67.1	64.6	58.4	53.4	49.5	(29)
		MCDB	54.8	56.7	62.3	71.4	81.1	87.4	94.3	93.2	84.5	69.4	60.2	56.2	(30)
	5%	WB	46.1	47.1	51.0	55.2	61.5	64.2	66.8	65.4	62.4	56.1	50.8	46.2	(31)
		MCDB	52.0	53.3	59.7	66.8	76.3	82.9	91.7	90.1	80.2	66.1	56.3	52.1	(32)
	10%	WB	43.9	44.9	49.1	52.8	59.3	62.1	65.1	64.0	60.5	54.1	48.2	43.5	(33)
		MCDB	49.3	50.7	57.5	62.9	71.9	79.1	89.1	87.2	77.4	63.2	53.8	48.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.8	16.9	19.2	21.0	23.0	26.7	25.6	23.1	18.2	12.6	10.2	(35)
		MCDBR	16.0	16.5	21.6	26.0	28.0	30.0	31.2	30.3	28.9	23.3	17.1	15.6	(36)
	5% WB	MCWBR	9.8	10.2	12.1	13.9	13.0	12.1	11.7	11.6	12.1	11.7	9.6	10.1	(37)
		MCWBR	15.3	15.7	19.8	24.8	25.5	28.0	29.1	28.0	25.9	20.7	14.9	15.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.289	0.294	0.305	0.334	0.352	0.353	0.354	0.357	0.337	0.317	0.294	0.285	(40)
		taud	2.528	2.537	2.537	2.440	2.412	2.434	2.441	2.423	2.462	2.523	2.541	2.513	(41)
	Ebn at Noon		263	285	296	293	290	289	287	282	279	271	258	251	(42)
		Edh at Noon	20	24	28	34	36	35	35	34	30	24	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		389	710	1109	1632	1986	2250	2430	2039	1506	931	488	315	(44)
	RadStd		44	88	87	107	139	121	87	114	94	72	42	36	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.97	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	-0.59	N/A	N/A	N/A	+162	N/A

Nomenclature: See separate page